

MAT 115E Introduction to Programming Language

Lab-2/ CRN : 23492

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1 Question 1

We want to perform a simple arithmetic operation on characters. Read two alphabetic characters from the user, the first one using **scanf()** and the second one using **getchar()**. Treat the characters as their ASCII values and compute their geometric mean using the formula below. Print the result with 7 digits after the decimal point.

$$\text{Geometric Mean} = \sqrt{x_1 \cdot x_2}$$

Example Scenario:

>Enter the first character: C
>Enter the second character: K
>Geometric mean of C and K is 70.8872343.

2 Question 2

Read two alphabetic characters from the user using the **getchar()** function. Find the character in the middle of these two characters you read and print it on the screen.

(Hint: you should employ the formula $\frac{\text{ch1} + \text{ch2}}{2}$, where ch1 and ch2 are two characters that you take from user.)

Example Scenario:

Example Scenario-1	Example Scenario-2
> Enter the first character: A	> Enter the first character: A
> Enter the second character: D	> Enter the second character: C
> Middle of these characters is B	> Middle of these characters is B

3 Question 3

Write a C program that accomplishes the following:

- Read a point in the format (x, y) from the user where x and y are two characters.
- Consider this point as a vector $\vec{v} = (x, y)$ originating from the origin.
- Compute the magnitude of the vector.

$$|\vec{v}| = \sqrt{x^2 + y^2}$$

- Print the result with 8 digits after the decimal point.

Example Scenario:

> Enter character point: (m, s)
> Magnitude of the vector: 158.44872987